François Roussel

PhD Student

francois.roussel@univ-lille.fr
in françois-roussel-281277332
FrancoisRoussel

PhD thesis in Signal Processing

Title	*Bayesian Estimate of Galaxy Clusters for Improved Cosmological Parameter Inference*
Location	*Centre de Recherche en Informatique, Signal et Automatique de Lille (CRIStAL, SigMA team)*, Villeneuve-d'Ascq, France
Supervisors	Jenny Sorce and Pierre-Antoine Thouvenin
Funding	ANR Chaire WILL UNIVERSITWINS Project hold by Jenny Sorce and Klaus Dolag
Keywords	Inverse problem solving, Bayesian hierarchical model, generative models, MCMC algorithms.

Research internships

Apr. 2025 - Sep. 2025 **Pre-PhD Research internship** *Centre de Recherche en Informatique, Signal et Automatique de Lille (CRIStAL, SigMA team)*, Villeneuve-d'Ascq, France

Title	Bayesian Estimate of Galaxy Clusters for Improved Cosmological Parameter Inference
Supervisors	Jenny Sorce and Pierre-Antoine Thouvenin
Funding	ANR Chaire IA SHERLOCK hold by Pierre Chainais
Keywords	Inverse problem solving, Bayesian inference, X-ray spectra modeling.

Apr. 2024 - Jul. 2024 **Research internship**, *Centre de Recherche en Informatique, Signal et Automatique de Lille (CRIStAL, SigMA team)*, Villeneuve-d'Ascq, France

Title	Galaxy cluster mass estimate using radial velocities
Supervisors	Pierre-Antoine Thouvenin and Jenny Sorce
Funding	ANR Chaire IA SHERLOCK hold by Pierre Chainais
Keywords	Inverse problem solving, physic based inference, Block Coordinate Forward Backward algorithm.

Research projects

Title	Mapping the distance between the simulated and the observed cosmic webs
Year	Master 2
Supervisors	Jérémie Boulanger and Jenny Sorce
Keywords	Constrained simulations, seperation metrics to define dissimilarities: likelihood, Mahalanobis and Sinkhorn distance.

Title	Galaxy cluster mass estimate using radial velocities
Year	Master 1
Supervisors	Pierre-Antoine Thouvenin and Jenny Sorce
Keywords	Inverse problem solving, pattern recognition, Gaussian Processes.

Title	Introduction to Topological Data Analysis (TDA)
Year	3rd y. Bachelor
Supervisors	Camille Sabbah
Keywords	Basic notions of topology, persistent homology, mapper algorithm, Betti number.

Education

2023 - 2025 **Master Degree in Data Science** (rank: 2/19), *Université de Lille*, Lille, France
Research-oriented Master entirely taught in english.

2020 - 2023 **Bachelor Degree in Applied Mathematics** (rank: 2/28), *Université de Lille*, Lille, France
Multidisciplinary bachelor focused on mathematics, economics and computer science.

Languages and programming Skills

Languages	French (Native), English (C1 level), German (B1 level)
Programming	Python, Git, R, LaTeX, SQL, HTML

Softwares: PowerBi, Excel

Experience in teaching

Apr. 2026 - Sep. 2026 **Co-supervision of two Master students** *Centre de Recherche en Informatique, Signal et Automatique de Lille (CRIStAL, SigMA team)*, Villeneuve-d'Ascq, France

Year 1st y. PhD Thesis

Topic Galaxy quenching, Virgo cluster, simulation-based inference

Keywords Assistance on supercomputers, cosmological simulations, RAMSES, RDRAMSES.

Oct. 2022 - Apr. 2023 **Mathematics Tutor of 1st y. Bachelor** *Mathématiques et informatique appliquées aux sciences humaines et sociales (MIASHS), Université de Lille*, Lille, France

Year 3rd y. Bachelor

Lectures Mathematical analysis and algebra

Keywords Limits, continuity, differentiation, differential equation, vector spaces, linear maps.

Apr. 2022 - Jul. 2022 **Intern in secondary education** *Collège Notre Dame de la Paix* and *Lycée Notre Dame de la Paix*, Lille, France

Year 2nd y. Bachelor

Lectures Small lectures in mathematics (from Year 7 to Year 12), a larger one at Year 13 (2×6 hours).

Keywords Artithmetics, geometry, statistics, probability, function analysis, law of large numbers.

Attendances

June, 11 2026 Bayesian inference for inverse problems, organized by J.B Courbot and P.-A. Thouvenin at *Institut Henri-Poincaré*, Paris, France.

June, 21 - 27 2026 20ème école d'été de Peyresq en Traitement du signal et des images : "modèles génératifs et transport optimal" (GRETSI), organized by A. Ferrari, G. Ginolhac, M. Massias and Q. Bertrand at Peyresq, France.